Calum Heldt @ calheldt.xyz

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EDUCATION

University of Nebraska at Omaha – 4.0 GPA Anticipated May '27

- Bachelor of Science in Mathematics with an Applied Math Concentration
- Minors in Computer Science, Education, and Sustainability

Graduate Coursework at Other Universities – 4.0 GPA

- Numerical Methods, University of Nebraska Lincoln — Spring 2026
- Differential Equations I-II, University of Nebraska Lincoln — Fall 2025 – Spring 2026
- Homological Algebra, University of Nebraska Lincoln — Fall 2025
- Local Fields, Preliminary Arizona Winter School (PAWS) — Fall 2024
- Modern Algebra I-II, University of Nebraska Lincoln — Fall 2024 – Spring 2025

AWARDS

- **ACCESS Discovery**: 1,500,000 credits to use on advanced computing resources, for fluid dynamics research co-authored with Prof. Larios (Sept. '26 – Sept. '27)
- **NASA Nebraska Space Grant**: $6,400 for research on Navier-Stokes-Voigt turbulence modeling (Sept. '26 – April '27)
- **Walter Scott, Jr. Scholarship:** the most prestigious scholarship extended by the University of Nebraska, covering full tuition, room, board, and more (Aug. '23 – May '27)
- **Regents Scholarship:** full tuition awarded to University of Nebraska students with an ACT score higher than 32, who maintain at least a 3.5 GPA (Aug. '23 – May '27)
- **Gilbert M. Hitchcock Memorial Scholarship in Mathematics:** $500 annually awarded to one promising incoming freshman math major (Aug. '23 – May '27)
- **James Earl Mathematics Endowed Scholarship:** $1,000 annually awarded to several outstanding math majors (Aug. '25, Aug. '26)
- **First place at Summer of Math Exposition (SoME)**: won the non-video entry category of Grant Anderson's (3Blue1Brown) annual SoME contest ('24)
- **NASA Nebraska Space Grant**: $3,000 for Galois Theory research, awarded to 30 Nebraska residents (Sept. '23 – April '24)
- **Success in math competitions:** 10 points on the Putnam exam ('23 & '24), 3rd statewide at UNL Math Day ('23 & '22), 10th as a team at Mu Alpha Theta Nationals ('23)

EXPERIENCE

Ecological Modeling with Boolean Networks Sept. 2025 – Present

- Determining the impact of biological and epidemiological factors on the spread of white-nose syndrome (WNS) in bats across the U.S., a catastrophic fungus-borne illness
- Working with Prof. Dora Velcsov and a peer, modeling spread of WNS with an open-source model I built from scratch in Python. Manuscript in progress.

Dynamical Systems Analysis of Slowly-Ramping Cahn-Hilliard June. 2026 – Aug. 2026

- Examining pattern selection of the non-autonomous Cahn-Hilliard equation using the phenomenological perspective of the Kibble-Zurek "freeze-out" mechanism
- Working with Prof. Ryan Goh at Boston University. Manuscript in progress.

TAMU Summer School Modeling and Simulation with PDEs May 2026

- Participated in rigorous lectures and small group projects with similarly advanced peers
- Simulated elasticity from scratch using the Navier-Lamé equations, derived from an Eulerian coordinate system, under Prof. Matthias Maier's supervision

Simulating the Navier-Stokes-Voigt Equations Sept. 2025 – Present

- Examining the Navier-Stokes-Voigt (NSV) equations used for fluid modeling
- Currently coding a second-order solver with the goal of analyzing statistics of NSV
- Working with Prof. Adam Larios at the University of Nebrska Lincoln

Simulating Liquid Crystals with Numerical PDEs May 2025 – Jan. 2026

- Created a repository for solving for a 2D system of liquid crystals at equilibrium
- Investigating formation of defects given various conditions from significant experiments
- Working with Prof. Andrew Hicks at Carnegie Mellon University

Modeling and Visualizing STEM Project June – Aug. 2024

- Created a public website at mavs.uno to make learning more accessible
- Intuitively explained STEM topics through interactive visuals and lessons
- Won Grant Sanderson's (3 Blue 1 Brown) Summer of Math Exposition for non-videos

Galois Theory Research June 2022 – May 2024

- Classified a family of Galois of order p^4 and p^6, with maximal nilpotency class
- Presented findings at the 2024 Hopf-Algebras and Galois Module Theory Conference
- Advised by Prof. Griff Elder at the University of Connecticut

Math Education Research June 2024 – April 2025

- Conduct research methods to examine college students' understanding of logarithms
- Use results to strengthen a conceptual framework for applying logarithmic identities
- Advised by Prof. Keith Gallagher at the University of Nebraska

Learning Assistant, Teaching Assistant, Tutor Aug. 2023 – Present

- Tutor students enrolled in intro proof writing courses
- Assist professors with classroom instruction and grading for various math courses

LEADERSHIP POSITIONS

MathCOUNTS Coach Aug. 2024 – Present

- Coach middle school students for the MathCOUNTS national competition series
- Teach problem-solving skills and promote math engagement

Math Club President May 2024 – May 2025

- Organize educational and social math events, prepare students for the Putnam Exam
- Foster a strong mathematical community on campus
- Previously held Secretary position and ran technical workshops

PEER-REVIEWED CONFERENCE PROCEEDINGS

27th Annual SIGMAA on RUME
All My Work is Right, Why Didn't I Get the Right Answer? When Logarithmic Identities Fail
Heldt, C. & Gallagher, K. 27th Annual SIGMAA on RUME Conference, Alexandria, VA, 2025.

ORAL PRESENTATIONS

MAA Sectional Meeting; UNO Research & Creative Activity Fair – Omaha April 2026
"Modeling liquid crystals with numerical PDEs"
Adapted previous summer research to educate a wider audience.
Hopf-Algebras and Galois Module Theory – Omaha May 2024
"Artin-Schreier presentations for Galois extensions of degree p^6 and maximal nilpotency class"
Presented to mathematicians from across the globe.

POSTER PRESENTATIONS

Joint Mathematics Meetings – Washington, D.C. Jan. 7th 2026
"Modeling liquid crystals with numerical PDEs"
Our reviewer said it was *"one of the higher levels of combined theoretical and applied mathematics on any poster [at the session]"*, with around 20 people visiting us in the 2-hour window.

COMPUTATIONAL SKILLS

Programming Languages

- Highly proficient: Python, MATLAB
- Some proficiency: command-line tools, C/C++, GAP

Markup Languages

- LaTeX, Beamer, CSS/SCSS, HTML, tikZ

Libraries

- Python: Firedrake, FEniCS
- Web Development: React, Ionic Angular

Applications

- GitHub, ParaView, VTK, Terminal

VOLUNTEERING

- Regularly giving time to the Completely KIDS nonprofit in Omaha. I've helped them implement a recycling program and re-examine their voluteering system in a new location (since Aug. '24)
- Helping run educational events for the Nebraska Mycological Society
- Attending Citzens' Climate Lobby and Nebraska Game and Parks recycling events in Omaha (since Aug. '24)
- Writing problems for and assisting at the University of Nebraska Omaha's Math Contest for high school students (since Aug. '23)

RELEVANT MATH COURSES

Currently Taking:

- Functional Analysis**
- Independent Study in Mathematical Fluid Dynamics

Completed:

- Graduate Differential Equations II* **(A)**
- Graduate Numerical Methods (Topics in Applied Math)* **(A+)**
- Graduate Boolean Networks** **(A+)**
- Applied Combinatorics **(A+)**
- Graduate Differential Equations I* **(A+)**
- Graduate Homological Algebra* **(A+)**
- Graduate Dynamical Systems and Chaos Theory** **(A+)**
- Finite Element Methods **(A+)**
- Graduate Modern Algebra II* **(A)**
- Partial Differential Equations **(A)**
- Graduate Local Fields ❋ **(A)**
- Measure Theory & Integration **(A+)**

Unless otherwise specified, these courses are/were at my home university, the University of Nebraska, Omaha (UNO). Since exhausting UNO's math catalog, I commute to the University of Nebraska, Lincoln (UNL) and take their graduate coursework.

* Taken at UNL with PhD-earning students
** Taken at UNO with Master's degree-earning students
❋ Taken online with the University of Connecticut, in conjunction with the Preliminary Arizona Winter School, hosted by Arizona State University

For more about me, please visit my website.